

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

Detecting overclaiming relative to guideline-supported evidence

Author: Ravi Bajaj | Target venue: ML4H 2026 Findings or Proceedings Track | Course: AI in Healthcare

Abstract

Large language models can draft fluent medication alerts that sound stronger than the evidence they cite. This paper narrows the capstone to one workflow: pharmacogenomic and genomic medication-alert text. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or extract evidence features from free text; it tests a deterministic controller over structured evidence annotations. In 30 author-designed synthetic cases, the controller blocked or narrowed all 20 designed overclaim archetypes, allowed all 10 bounded alerts, and produced no inappropriate denials. Action conformance was 30/30, while primary-gate conformance was 28/30 because citation-first precedence changed the recorded primary gate for PGX19 and PGX24. These counts demonstrate specification conformance against the author-designed case bank, not independent clinical safety, diagnostic accuracy, generalization, or patient benefit.

1. Introduction

The prior capstone package showed that a large set of mathematical and assurance ideas could be connected to medical LLM evaluation. The reviewer-facing problem is that breadth weakened the causal chain. This revision makes pruning the method: one workflow, one failure mode, three gates, and a measurable evaluation plan.

The chosen workflow is pharmacogenomic or genomic medication-alert text. It is attractive because guideline resources such as CPIC and ClinPGx already distinguish genotype, phenotype, evidence level, and prescribing recommendation. That creates a concrete setting where an LLM can be audited for a specific error: converting real but limited evidence into a stronger clinical claim.

2. Related Work and Evidence Boundary

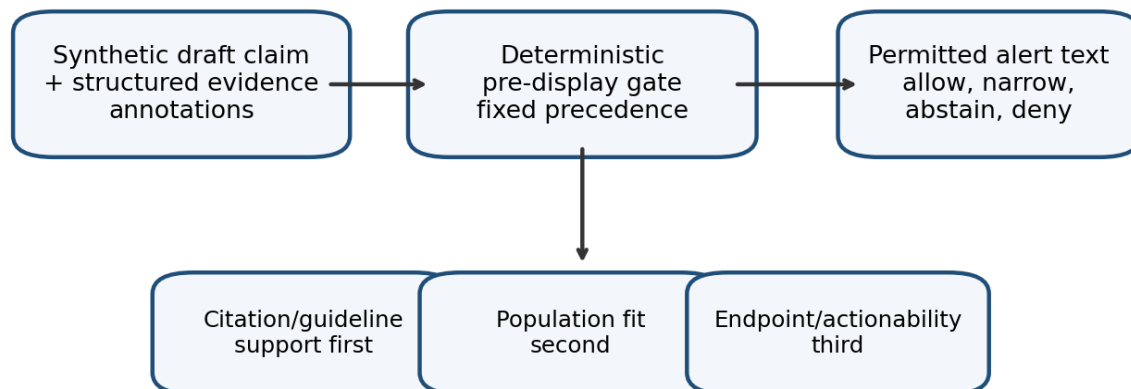
The course theme from Byrne, Domenico, and Moore is that patient-outcome claims for healthcare AI require pragmatic clinical evaluation. Stegenga's Medical Nihilism motivates caution when evidentiary pipelines make interventions appear stronger than they are. FDA-NIH BEST materials provide the endpoint vocabulary: biomarkers, surrogate endpoints, validated surrogates, and clinical outcomes are not interchangeable.

Pharmacogenomics adds a useful positive control. CPIC-style resources often support bounded prescribing language, such as avoiding a drug in a specific genotype/phenotype context or reviewing dose because of toxicity risk. Those resources do not automatically support claims that an LLM improves outcomes, that an association is causal, that a finding transports across populations without recalibration, or that a weak genetic association authorizes treatment denial.

3. Methods

The controller is a deterministic pre-display gate. It is not presented as an autonomous clinical agent. In Stage 1, it receives a synthetic candidate alert plus structured evidence annotations assigned in the case bank; it does not parse the draft_claim text, call an LLM, verify citations, or extract EHR/population-fit features. Those extraction tasks are the central Stage 2 dependency. The fixed precedence order is citation/guideline support first,

population fit second, and endpoint/actionability third. The controller selects one of five permission levels: allow bounded alert, narrow claim, abstain for population fit, abstain for citation/guideline support, or deny unsupported action.



Stage 1 does not call an LLM or extract evidence features from free text.

Figure 1. Pruned architecture for the evidence gate.

Gate	Question	Permitted effect
Citation/guideline support	Is the cited guideline real, current, relevant, and strong enough for the claim?	Deny fabricated claims; abstain on uncertain/conflicting support.
Population fit	Does source evidence plausibly fit the target patient/cohort?	Require source-specific validation or abstain from transport.
Endpoint/actionability	Does the evidence support medication action, only surrogate/process language, or no action?	Allow bounded alert, narrow wording, or deny unsupported action.

4. Stage 1 Specification-Conformance Scaffold

The current case bank contains 30 author-designed synthetic cases. Ten are bounded aligned alerts; twenty intentionally contain overclaim archetypes involving universal action language, unsupported genetics, population mismatch, citation/guideline weakness, variant uncertainty, or obsolete/context-shifted evidence. The 20/10 split is a stress-test design choice, not a measured clinical base rate. Each case has an expected action and expected primary gate, but the controller does not read either expected label.

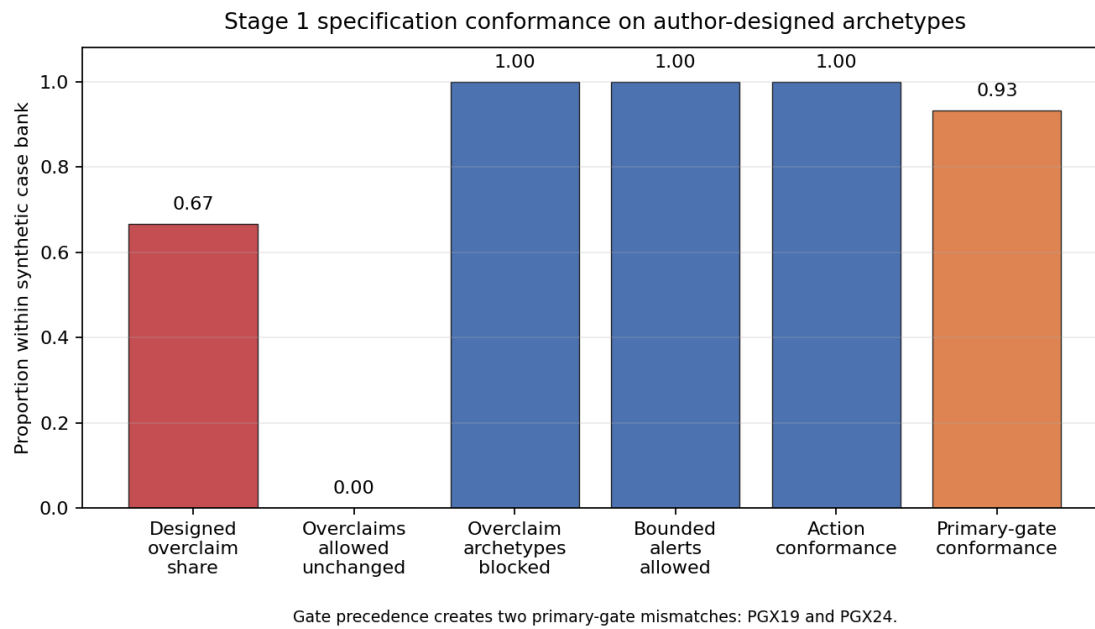
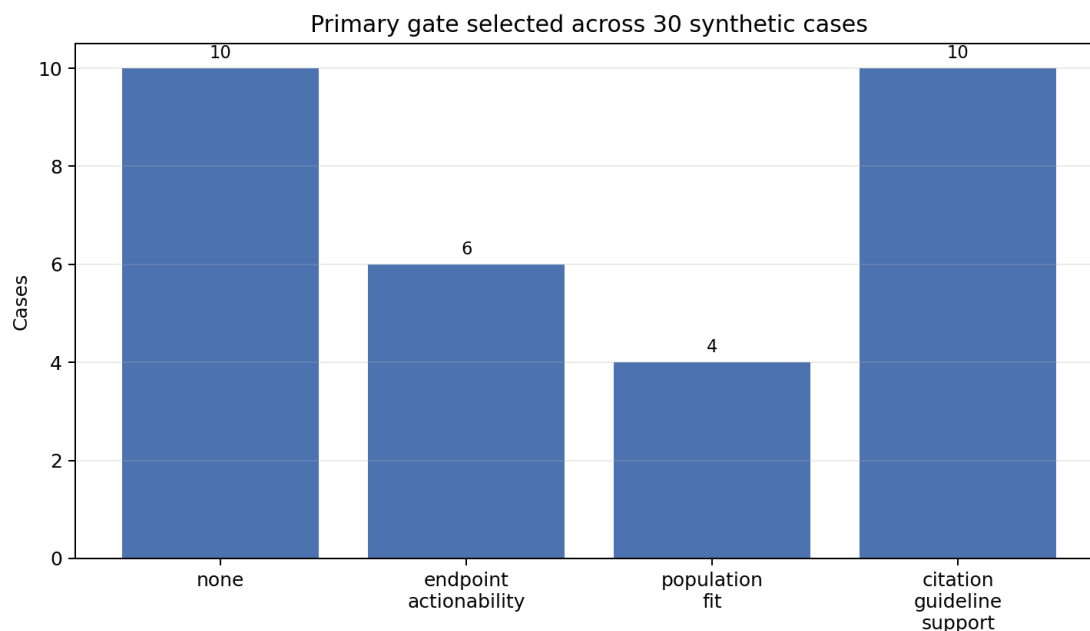


Figure 2. Stage 1 specification-conformance counts on author-designed synthetic archetypes.

Result	Value	Claim allowed
Author-designed overclaim share	20/30	Stress-test case mix, not a measured clinical base rate.
Overclaim archetypes allowed unchanged	0/20	Derived as overclaim cases that reached allow bounded alert unchanged.
Designed overclaim archetypes blocked	20/20	Controller routes intentionally overstrong claims to narrow, abstain, or deny.
Bounded alerts allowed	10/10	Controller permits aligned bounded alerts in the synthetic case bank.
Action conformance	30/30	Actual action matched expected action for all cases.
Primary-gate conformance	28/30	PGX19 and PGX24 differ because citation support fires before endpoint or population fit.
Inappropriate denial	0/10	No aligned bounded alerts were denied too strongly in Stage 1.



Error analysis: PGX19 and PGX24 are action-level successes but primary-gate mismatches. In both cases, citation/guideline support fires before the gate that motivated the case design. This exposes the controller precedence order and prevents the Stage 1 result from being presented as a perfect gate-identification result.

Figure 3. Primary gate selected by the deterministic controller.

5. Planned Stage 2

Stage 2 is the part that would make the paper clinically interpretable. It should compare ungated LLM drafts with gated drafts on independently authored cases. Reviewers should be blinded to arm when feasible and should classify overclaiming, inappropriate denial, preserved clinician authority, population-fit concerns, citation support, and error category.

- Primary outcome: reduction in unsupported evidence overclaiming.
- Safety outcome: inappropriate denial of bounded, guideline-supported alerts.
- Secondary outcomes: calibration of overclaim risk, interrater agreement, and qualitative error taxonomy.
- Boundary: no real patient data are needed for Stage 1; any Stage 2 use of clinical cases requires appropriate governance or synthetic case authoring.

6. Discussion

The key lesson is that an LLM does not need to be proven clinically useful before it can be disciplined. A pre-display gate can enforce a smaller claim grammar: a guideline supports a bounded alert, a population mismatch forces abstention, and an unverifiable citation carries no medical authority. This is not a substitute for pragmatic trials; it is a way to prevent a paper, product, or alert from claiming more than its evidence can bear.

The most important limitation is that the present results are synthetic and author-designed. They demonstrate that the implementation follows the intended action rules, not that the system generalizes. A second limitation is that the hardest upstream task is assumed rather than solved: the case bank supplies structured evidence annotations instead of extracting them from live LLM output, citations, or EHR context.

7. Conclusion

A cohesive capstone can be built from the original implementation by pruning. The revised project studies one practical genetics workflow and evaluates whether a three-gate controller can discipline LLM alert text so it does not inflate guideline-supported evidence into unsupported clinical authority.

References and Web Sources

- ML4H 2026 home. <https://ml4h.ahli.cc/>
- ML4H 2026 call for participation. <https://ml4h.ahli.cc/submit/call-for-papers/>
- Byrne, D. W., Domenico, H. J., & Moore, R. P. (2024). Artificial intelligence for improved patient outcomes: The pragmatic randomized controlled trial is the secret sauce.. <https://kjronline.org/DOIx.php?id=10.3348%2Fkjr.2023.1016>
- Stegenga, J. (2018). Medical Nihilism. Oxford University Press.. <https://global.oup.com/academic/product/medical-nihilism-9780198747048>
- FDA-NIH BEST Resource. <https://www.ncbi.nlm.nih.gov/books/NBK326791/>
- CPIC guidelines overview. <https://cpicpgx.org/guidelines/>
- ClinPGx CPIC CYP2C19-clopidogrel guideline. <https://www.clinpgx.org/guideline/PA166251443>
- CPIC CYP2C19-clopidogrel 2022 update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9287492/>
- CPIC DPYD-fluoropyrimidines guideline. <https://www.clinpgx.org/guideline/PA166251462>
- CPIC DPYD-fluoropyrimidines update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5760397/>
- CPIC TPMT/NUDT15-thiopurines update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12997511/>
- ClinPGx HLA-B-abacavir guideline. <https://www.clinpgx.org/guideline/PA166251444>
- ClinPGx HLA-A/HLA-B-carbamazepine guideline. <https://www.clinpgx.org/guideline/PA166251448>
- ClinVar. <https://www.ncbi.nlm.nih.gov/clinvar/>
- gnomAD. <https://gnomad.broadinstitute.org/>
- All of Us Research Program. <https://allofus.nih.gov/>
- Genome India project. <https://genomeindia.in/>
- Psifas / Mosaic. <https://partnership.psifas.org.il/>